Miter Bend Calculation Report

ASME B31.1-2022 | Pressure path and segment wall

DESIGN EQUATION

ASME B31.1-2022, 104.2.3(a)-(c)
Simplified paths: 10 psi or 100 psi + all applicable conditions
Otherwise: qualification under 104.7
(c)(3), close: ts = tm (2-r/R)/[2(1-r/R)]
(c)(3), wide: ts = tm [1 + 0.64 sqrt(r/ts) tan(theta)]
Segment thickness does not replace qualification.
Pressure equations: P and S in MPa; dimensions in mm; input bar x 0.1 = MPa.

Project	QA miter
Prepared by	Device preview
Report date	2026-10-06 10:57

01 INPUTS

Design gauge pressure	8 bar
Miter cut angle theta	22.5 deg
Nominal wall tn / crotch B	6.02 / 40 mm

02 DECLARED CONDITIONS AND PATH

(b)(1)-(2) θ ≤ 22.5° and every segment has B ≥ 6tn: Declared met

(a)(2) each segment thickness ≥ that required by para. 104.1: Declared met

(a)(3) fluid nonflammable, nontoxic and incompressible: Declared met

(a)(4) fewer than 7,000 full pressure cycles: Declared met

(a)(5) full penetration welds: Declared met

Selected path	§104.2.3(c) — qualification per §104.7 required
Pressure path limit	Not a simplified path

QUALIFICATION REQUIRED under 104.7. No approved pressure verdict is issued.

03 SEGMENT THICKNESS

Declared spacing	CLOSE
Straight-pipe minimum tm	5 mm
Pipe radius r	50 mm
Bend radius R	150 mm
Required segment wall ts	6.25 mm

104.2.3(c)(3) wall calculation does not replace qualification under 104.7. Geometry and spacing are designer declarations using ASME B31J.

04 REFERENCES AND LIMITS

Sources: B31.3-2018 304.2.3, supplied printed pp.22-23; B31.1-2022 104.2.3, supplied printed p.29. Inputs are user supplied. No external-pressure, fatigue, remaining-life or complete code qualification is performed.